

CoRoT-1b

R-2540

Benchmark run · workflow W8-benchmark-deep · record deep-bench_CoRoT-1_200221 · created 2026-08-26T00:01:04+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458900.6393 +/- 0.0044 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.052 +/- 0.025
Transit depth (Rp/Rs)^2	0.27 +/- 0.26 [%]
Orbital Inclination (inc)	80.12 +/- 2.88
Airmass coefficient 1 (a1)	0.612 +/- 0.011
Airmass coefficient 2 (a2)	0.399 +/- 0.058
Scatter in the residuals of the lightcurve fit is	1.8 %
Best Comparison Star	#2 - [83, 306]
Optimal Method	PSF photometry
Transit Duration (day)	0.051 +/- 0.028

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.052 ± 0.025 vs 0.1388 (deviation 2.78σ)
Duration fit vs published	0.051 d vs 0.1046 d (deviation 1.53σ)
Mid-time O-C	9.9 min
Depth SNR	1.7
Residual scatter	1.8 %

Light curve

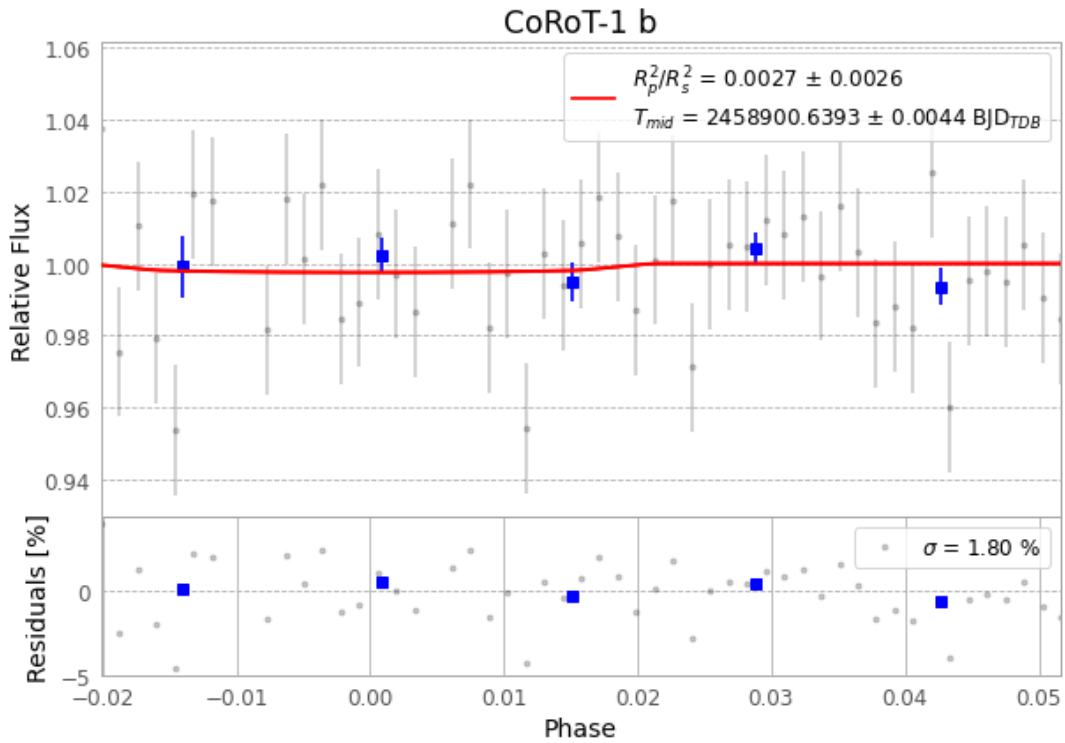


Figure 1 — FinalLightCurve_CoRoT-1 b_2020-02-20.png (SHA-256 a25671f7bef250d4...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [79, 355], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 99720266c706b157...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

53 FITS frames (35 MB), CoRoT-1200221023012.FITS → CoRoT-1200221050612.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-1-200221.log (2a36e83fd8dd...), FinalLightCurve_CoRoT-1 b_2020-02-20.png (a25671f7bef2...), FinalLightCurve_CoRoT-1 b_2020-02-20.csv (7aae5515767b...)

Compute resources

Wall time 123.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-26 00:01Z.